Project Title

Word-Level Plagiarism Checker for Short Documents

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT
Multi-Document Plagiarism Detection and Similarity Analysis System is a Python-based application that detects textual similarities across multiple documents using advanced data structures and algorithms. The system allows users to upload documents in various formats, preprocess the text through tokenization, stop-word removal, and stemming, and perform efficient pattern matching using Knuth-Morris-Pratt (KMP) and Rabin-Karp algorithms. To improve search efficiency for large document collections, the system utilizes suffix-based indexing and hash-based data structures. Similarity between documents is computed using Jaccard Similarity and TF-IDF with Cosine Similarity, while dynamic programming techniques are employed to identify the longest common consequences and optimize phrase matching. The application generates similarity reports, ranks the most similar documents, and highlights matching content. A configurable similarity threshold alerts users to potential plagiarism and promotes academic integrity. The project demonstrates the practical application of advanced string processing, dynamic programming, indexing techniques, and algorithm optimization for large-scale document analysis.

Introduction
Plagiarism detection has become increasingly important in educational institutions and research organizations due to the widespread availability of digital content. Traditional manual methods of comparing documents are time-consuming and inefficient, especially when handling large numbers of files. Existing plagiarism detection tools such as Turnitin and iThenticate use advanced text comparison techniques to identify similarities between documents. However, these tools are commercial, expensive, and operate as closed systems, making it difficult to understand the algorithms behind the similarity calculations.
Most existing approaches rely on exact text matching or predefined databases and provide limited insight into the underlying data structures and algorithms. They also offer little flexibility for educational purposes where algorithm implementation is a key learning objective.
The proposed Multi-Document Word-Level Plagiarism Detection and Similarity Analysis System addresses these limitations by implementing advanced DSA concepts such as KMP, Rabin-Karp, Hash Maps, Inverted Indexing, and Dynamic Programming for efficient document comparison. The system compares multiple uploaded documents, calculates similarity percentages, highlights matching content, ranks similar documents, and generates plagiarism reports with configurable warning thresholds, providing both an effective plagiarism detection tool and a practical application of advanced algorithms.

Software and hardware requirements:

Software Requirements:-
- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.10+
- IDE: Visual Studio Code or PyCharm

- Libraries: NLTK, PyPDF2, python-docx, scikit-learn, NumPy, pandas
- Database: SQLite/MySQL (optional)

Hardware Requirements:-

- Processor: Intel Core i3 / AMD Ryzen 3 or higher
- RAM: 4 GB (8 GB recommended)
- Storage: 10 GB free disk space
- Display: 1366 × 768 or higher
- Internet: Required for installing libraries and optional online document comparison.

Signature of the Guide Course Instructor